

# TensaiDevs

## Student Reporting and Performance Tracking System

### API Reference Documentation

Version 1.0 | Backend: Node.js + Express | Storage: JSON Files

## 1. Overview

The TensaiDevs API is a RESTful backend service built with Node.js and Express. It powers the student reporting and performance tracking system for TensaiDevs. All data is stored in flat JSON files located in `src/data/`. The API is designed to be migrated to a database at a later stage with minimal changes.

### 1.1 Base URL

```
http://localhost:3000
```

### 1.2 Authentication

The API uses JSON Web Tokens (JWT) for authentication. To access protected endpoints, the client must include the token in the Authorization header of every request.

```
Authorization: Bearer <your_jwt_token>
```

Tokens are issued on successful login and expire after 7 days. All endpoints that require authentication return 401 Unauthorized if no valid token is provided.

### 1.3 User Roles

| Role    | Description                                                                                                                |
|---------|----------------------------------------------------------------------------------------------------------------------------|
| admin   | Full access to all endpoints. Can manage students, reports, evaluations, tags, attendance, analytics, and system settings. |
| student | Restricted access. Can only submit and view their own reports, view their own analytics, and view their own attendance.    |

### 1.4 Standard Response Format

Every response follows this consistent structure:

```
{
  "success": true | false,
  "message": "Human readable message",
  "data": { ... }
}
```

On validation errors, an additional errors array is included:

```
{
  "success": false,
  "message": "Validation failed",
  "errors": [{ "field": "email", "msg": "Valid email is required" }]
}
```

## 1.5 HTTP Status Codes

| Code | Meaning               | When it occurs                                                          |
|------|-----------------------|-------------------------------------------------------------------------|
| 200  | OK                    | Successful GET, PUT, PATCH requests                                     |
| 201  | Created               | Successful POST requests that create a new resource                     |
| 400  | Bad Request           | Missing required fields, validation failure, or business rule violation |
| 401  | Unauthorized          | No token provided, token expired, or token is invalid                   |
| 403  | Forbidden             | Valid token but the user role does not have permission                  |
| 404  | Not Found             | The requested resource does not exist                                   |
| 500  | Internal Server Error | Unexpected server-side error                                            |

## 1.6 Seed Data

The system ships with the following pre-seeded accounts for testing:

| Role    | Email                     | Password   | IDs                                              |
|---------|---------------------------|------------|--------------------------------------------------|
| Admin   | hassansattar147@gmail.com | admin123   | admin-001                                        |
| Student | ahmad.gul@tensaidevs.com  | student123 | student-001 / batch-001 / course-001 / group-001 |
| Student | muhib@tensaidevs.com      | student123 | student-002 / batch-001 / course-002 / group-003 |

## 2. Authentication Endpoints

### 2.1 Login

|                                                                                                                                            |                            |
|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>POST</b>                                                                                                                                | <b>/api/auth/login</b>     |
| Authenticates a user using email and password. Returns a JWT token and user profile on success. Works for both admin and student accounts. |                            |
| <b>Auth</b>                                                                                                                                | No authentication required |

#### Request Body

| Field    | Type   | Required | Example                   | Description                            |
|----------|--------|----------|---------------------------|----------------------------------------|
| email    | string | Yes      | hassansattar147@gmail.com | Registered email address of the user   |
| password | string | Yes      | admin123                  | Account password. Minimum 6 characters |

#### Example Request Body

```
{ "email": "hassansattar147@gmail.com", "password": "admin123" }
```

#### Success Response (200)

```
{
  "success": true,
  "message": "Login successful",
  "data": {
    "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
    "user": { "id": "admin-001", "name": "Hassan Sattar", "email": "...", "role": "admin" }
  }
}
```

#### Error Response (401)

```
{ "success": false, "message": "Invalid email or password" }
```

### 2.2 Get My Profile

|                                                                                         |                               |
|-----------------------------------------------------------------------------------------|-------------------------------|
| <b>GET</b>                                                                              | <b>/api/auth/me</b>           |
| Returns the full profile of the currently authenticated user. No request body required. |                               |
| <b>Auth</b>                                                                             | Requires JWT token (any role) |

#### Success Response (200)

```
{
  "success": true,
```

```
"data": { "user": { "id": "admin-001", "name": "Hassan Sattar", "role": "admin",  
"isActive": true, ... } }  
}
```

## 2.3 Change Password

|                                                                                                                       |                                  |
|-----------------------------------------------------------------------------------------------------------------------|----------------------------------|
| <b>PUT</b>                                                                                                            | <b>/api/auth/change-password</b> |
| Allows the currently authenticated user to change their own password. Requires the current password for verification. |                                  |
| <b>Auth</b>                                                                                                           | Requires JWT token (any role)    |

### Request Body

| Field           | Type   | Required | Example    | Description                                |
|-----------------|--------|----------|------------|--------------------------------------------|
| currentPassword | string | Yes      | admin123   | The user current password for verification |
| newPassword     | string | Yes      | newpass456 | The new password.<br>Minimum 6 characters  |

## 2.4 Admin Reset Student Password

|                                                                                               |                                            |
|-----------------------------------------------------------------------------------------------|--------------------------------------------|
| <b>PUT</b>                                                                                    | <b>/api/auth/reset-password/:studentId</b> |
| Admin resets the password of any student account. The studentId is passed as a URL parameter. |                                            |
| <b>Auth</b>                                                                                   | Requires JWT token — Admin only            |

### URL Parameters

| Field     | Type         | Required | Example     | Description                                                |
|-----------|--------------|----------|-------------|------------------------------------------------------------|
| studentId | string (URL) | Yes      | student-001 | The unique ID of the student whose password is being reset |

### Request Body

| Field       | Type   | Required | Example    | Description                                      |
|-------------|--------|----------|------------|--------------------------------------------------|
| newPassword | string | Yes      | newpass456 | The new password to set.<br>Minimum 6 characters |

### 3. Settings Endpoints

#### 3.1 Get System Settings

|                                                                                                                |                               |
|----------------------------------------------------------------------------------------------------------------|-------------------------------|
| GET                                                                                                            | /api/settings                 |
| Returns current system-wide settings including the daily submission deadline and maximum late submission days. |                               |
| Auth                                                                                                           | Requires JWT token (any role) |

Success Response (200)

```
{
  "success": true,
  "data": {
    "settings": {
      "submissionDeadline": "23:00",
      "submissionDeadlineUpdatedAt": "2025-01-01T00:00:00.000Z",
      "maxLateSubmissionDays": 3
    }
  }
}
```

#### 3.2 Update Submission Deadline

|                                                                                                                                    |                                 |
|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| PUT                                                                                                                                | /api/settings/deadline          |
| Admin sets the daily report submission deadline. Must be in HH:MM 24-hour format. The new deadline takes effect the following day. |                                 |
| Auth                                                                                                                               | Requires JWT token — Admin only |

Request Body

| Field              | Type   | Required | Example | Description                                                            |
|--------------------|--------|----------|---------|------------------------------------------------------------------------|
| submissionDeadline | string | Yes      | 22:30   | New deadline in HH:MM 24-hour format. For example 23:00 means 11:00 PM |

## 4. Student Management Endpoints

All student management endpoints are restricted to admin role only.

### 4.1 List Students

|                                                                                                         |                                 |
|---------------------------------------------------------------------------------------------------------|---------------------------------|
| GET                                                                                                     | /api/students                   |
| Returns a paginated list of all student accounts. Supports multiple filter parameters via query string. |                                 |
| Auth                                                                                                    | Requires JWT token — Admin only |

Query Parameters (all optional)

| Field    | Type    | Required | Example        | Description                                              |
|----------|---------|----------|----------------|----------------------------------------------------------|
| name     | string  | No       | Ahmad          | Filter by student name (partial match, case-insensitive) |
| batchId  | string  | No       | batch-001      | Filter by batch ID                                       |
| courseId | string  | No       | course-001     | Filter by course ID                                      |
| groupId  | string  | No       | group-001      | Filter by group ID                                       |
| tag      | string  | No       | High Performer | Filter by tag name (exact match, case-insensitive)       |
| isActive | boolean | No       | true           | Filter by active status. Accepts true or false as string |
| page     | integer | No       | 1              | Page number for pagination. Defaults to 1                |
| limit    | integer | No       | 20             | Number of results per page. Defaults to 20               |

Success Response (200)

```
{
  "success": true,
  "data": {
    "students": [
      { "id": "student-001", "name": "Ahmad Gul", "email": "...", "role": "student",
        "batchId": "batch-001", "courseId": "course-001", "groupId": "group-001",
        "isActive": true, "tags": [{ "id": "tag-001", "name": "High Performer" }] }
    ],
    "total": 2, "page": 1, "limit": 20
  }
}
```

## 4.2 Get Student by ID

|                                                                      |                                 |
|----------------------------------------------------------------------|---------------------------------|
| <b>GET</b>                                                           | <b>/api/students/:studentId</b> |
| Returns a single student's full profile including all assigned tags. |                                 |
| <b>Auth</b>                                                          | Requires JWT token — Admin only |

### URL Parameters

| Field     | Type   | Required | Example     | Description                  |
|-----------|--------|----------|-------------|------------------------------|
| studentId | string | Yes      | student-001 | The unique ID of the student |

## 4.3 Create Student

|                                                                                                                                                                                                             |                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| <b>POST</b>                                                                                                                                                                                                 | <b>/api/students</b>            |
| Admin creates a new student account. The student is assigned to a specific batch, course, and group during creation. The course must belong to the batch and the group must belong to the batch and course. |                                 |
| <b>Auth</b>                                                                                                                                                                                                 | Requires JWT token — Admin only |

### Request Body

| Field    | Type   | Required | Example             | Description                                          |
|----------|--------|----------|---------------------|------------------------------------------------------|
| name     | string | Yes      | Test Student        | Full name of the student                             |
| email    | string | Yes      | test@tensaidevs.com | Email address used for login. Must be unique         |
| password | string | Yes      | student123          | Initial password. Minimum 6 characters               |
| batchId  | string | Yes      | batch-001           | ID of the batch to enroll the student in             |
| courseId | string | Yes      | course-001          | ID of the course. Must be assigned to the batch      |
| groupId  | string | Yes      | group-001           | ID of the group. Must belong to the batch and course |

## 4.4 Update Student

|                                                                                                                                                                                                           |                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| <b>PUT</b>                                                                                                                                                                                                | <b>/api/students/:studentId</b> |
| Admin updates a student's name or reassigns them to a different batch, course, or group. All body fields are optional. If any of batchId, courseId, or groupId is provided, the combination is validated. |                                 |
| <b>Auth</b>                                                                                                                                                                                               | Requires JWT token — Admin only |

### URL Parameters

| Field     | Type   | Required | Example     | Description                            |
|-----------|--------|----------|-------------|----------------------------------------|
| studentId | string | Yes      | student-001 | The unique ID of the student to update |

**Request Body (all optional)**

| Field    | Type   | Required | Example       | Description                                               |
|----------|--------|----------|---------------|-----------------------------------------------------------|
| name     | string | No       | Ahmad Updated | New display name for the student                          |
| batchId  | string | No       | batch-002     | New batch assignment                                      |
| courseId | string | No       | course-002    | New course assignment. Must belong to the batch           |
| groupId  | string | No       | group-003     | New group assignment. Must belong to the batch and course |

## 4.5 Toggle Student Active Status

|                                                                                                                                                                          |                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <b>PATCH</b>                                                                                                                                                             | <b>/api/students/:studentId/toggle-status</b> |
| Toggles the isActive status of a student. Deactivated students cannot log in but all their reports, attendance, and evaluations are preserved. No request body required. |                                               |
| <b>Auth</b>                                                                                                                                                              | Requires JWT token — Admin only               |

**URL Parameters**

| Field     | Type   | Required | Example     | Description                  |
|-----------|--------|----------|-------------|------------------------------|
| studentId | string | Yes      | student-001 | The unique ID of the student |



## 5. Program Structure Endpoints

*Batches, courses, and groups form the enrollment hierarchy. These endpoints are all admin only.*

### 5.1 List Batches

|                                                                            |                                 |
|----------------------------------------------------------------------------|---------------------------------|
| <b>GET</b>                                                                 | <b>/api/program/batches</b>     |
| Returns all batches. Each batch includes a list of courses assigned to it. |                                 |
| <b>Auth</b>                                                                | Requires JWT token — Admin only |

### 5.2 Get Batch by ID

|                                   |                                      |
|-----------------------------------|--------------------------------------|
| <b>GET</b>                        | <b>/api/program/batches/:batchId</b> |
| Returns a single batch by its ID. |                                      |
| <b>Auth</b>                       | Requires JWT token — Admin only      |

#### URL Parameters

| Field   | Type   | Required | Example   | Description                |
|---------|--------|----------|-----------|----------------------------|
| batchId | string | Yes      | batch-001 | The unique ID of the batch |

### 5.3 Create Batch

|                                                                                                                |                                 |
|----------------------------------------------------------------------------------------------------------------|---------------------------------|
| <b>POST</b>                                                                                                    | <b>/api/program/batches</b>     |
| Creates a new batch. Optionally copies course assignments from a previous batch by providing carryCoursesFrom. |                                 |
| <b>Auth</b>                                                                                                    | Requires JWT token — Admin only |

#### Request Body

| Field            | Type   | Required | Example       | Description                                                                 |
|------------------|--------|----------|---------------|-----------------------------------------------------------------------------|
| name             | string | Yes      | Batch 2       | Unique name for the batch. Duplicate names are rejected                     |
| description      | string | No       | Second intake | Optional description of the batch                                           |
| carryCoursesFrom | string | No       | batch-001     | If provided, copies all course assignments from that batch to the new batch |

### 5.4 Assign Course to Batch

**POST****/api/program/batches/:batchId/courses**

Assigns an existing course to a specific batch. Once assigned, students can be enrolled in that course within that batch.

**Auth**

Requires JWT token — Admin only

**URL Parameters**

| Field   | Type   | Required | Example   | Description                       |
|---------|--------|----------|-----------|-----------------------------------|
| batchId | string | Yes      | batch-002 | The batch to assign the course to |

**Request Body**

| Field    | Type   | Required | Example    | Description                    |
|----------|--------|----------|------------|--------------------------------|
| courseId | string | Yes      | course-001 | The ID of the course to assign |

## 5.5 List Courses

**GET****/api/program/courses**

Returns all courses defined in the system regardless of which batches they are assigned to.

**Auth**

Requires JWT token — Admin only

## 5.6 Create Course

**POST****/api/program/courses**

Creates a new course. Domain must be one of the four permitted values.

**Auth**

Requires JWT token — Admin only

**Request Body**

| Field       | Type   | Required | Example                           | Description                                                                |
|-------------|--------|----------|-----------------------------------|----------------------------------------------------------------------------|
| name        | string | Yes      | Digital Marketing                 | Unique course name                                                         |
| description | string | No       | Content and social media strategy | Optional description                                                       |
| domain      | string | Yes      | professional_skills               | Must be one of: programming, designing, video_editing, professional_skills |

## 5.7 List Groups

**GET****/api/program/groups**

Returns all groups. Filter by batchId and/or courseId using query parameters.

**Auth** Requires JWT token — Admin only

#### Query Parameters (all optional)

| Field    | Type   | Required | Example    | Description             |
|----------|--------|----------|------------|-------------------------|
| batchId  | string | No       | batch-001  | Filter groups by batch  |
| courseId | string | No       | course-001 | Filter groups by course |

## 5.8 Create Group

**POST** /api/program/groups

Creates a new group inside a specific batch and course combination. The course must already be assigned to the batch.

**Auth** Requires JWT token — Admin only

#### Request Body

| Field    | Type   | Required | Example    | Description                                                     |
|----------|--------|----------|------------|-----------------------------------------------------------------|
| name     | string | Yes      | Group C    | Group name. Must be unique within the same batch and course     |
| batchId  | string | Yes      | batch-001  | The batch this group belongs to                                 |
| courseId | string | Yes      | course-001 | The course this group belongs to. Must be assigned to the batch |

## 6. Report Endpoints

### 6.1 Report Status Values

| Status         | Description                                                                  |
|----------------|------------------------------------------------------------------------------|
| draft          | Saved by the student but not yet submitted. Not visible to admins.           |
| pending        | Submitted by the student and awaiting admin review.                          |
| needs_revision | Admin has reviewed and requires the student to make changes before approval. |
| approved       | Admin has reviewed and accepted the report.                                  |
| rejected       | Admin has reviewed and determined the report does not meet expectations.     |

### 6.2 Submit or Save Draft Report (Student)

|                                                                                                                                                                                                                               |                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| POST                                                                                                                                                                                                                          | /api/reports                      |
| Student submits a daily report or saves it as a draft. If a report for the same date already exists it is updated. Reports can be submitted for up to 3 days in the past. All text fields are required when isDraft is false. |                                   |
| Auth                                                                                                                                                                                                                          | Requires JWT token — Student only |

#### Business Rules

| Field                  | Type                                                         | Required | Example | Description |
|------------------------|--------------------------------------------------------------|----------|---------|-------------|
| Rule                   | Detail                                                       |          |         |             |
| One report per day     | A student can only have one report entry per calendar date   |          |         |             |
| Late submission window | Reports can be submitted for up to 3 past calendar days      |          |         |             |
| Draft edit window      | Drafts can be edited any time before submission              |          |         |             |
| Submitted report edit  | Editable until the admin reviews it or the deadline passes   |          |         |             |
| Needs revision flow    | When admin marks needs_revision, student edits and resubmits |          |         |             |

|  |                               |  |  |  |
|--|-------------------------------|--|--|--|
|  | (status goes back to pending) |  |  |  |
|--|-------------------------------|--|--|--|

**Request Body**

| Field         | Type          | Required | Example                             | Description                                                          |
|---------------|---------------|----------|-------------------------------------|----------------------------------------------------------------------|
| reportDate    | string (date) | Yes      | 2026-05-18                          | Date the report covers in YYYY-MM-DD format. Cannot be a future date |
| whatStudied   | string        | Yes*     | JavaScript promises                 | Topics or skills studied. Required when isDraft is false             |
| timeSpent     | string        | Yes*     | 3 hours                             | Total time spent studying. Free text. Required when isDraft is false |
| whatLearned   | string        | Yes*     | How to chain promises               | Key takeaways or new understanding. Required when isDraft is false   |
| challenges    | string        | Yes*     | Event loop was confusing            | Difficulties or blockers encountered. Required when isDraft is false |
| progressNotes | string        | Yes*     | Will review error handling tomorrow | Additional context or progress notes. Required when isDraft is false |
| isDraft       | boolean       | No       | false                               | Set to true to save as draft without submitting. Defaults to false   |

**Success Response (201 for new, 200 for update)**

```
{
  "success": true,
  "message": "Report submitted successfully",
  "data": {
    "report": {
      "id": "report-abc123", "studentId": "student-001",
      "reportDate": "2025-05-18", "status": "pending",
      "isLate": false, "submittedAt": "2025-05-18T19:30:00.000Z",
      "whatStudied": "...", "timeSpent": "...", "whatLearned": "...",
      "challenges": "...", "progressNotes": "...",
      "evaluation": null
    }
  }
}
```

## 6.3 Get My Reports (Student)

|                                                                                                              |                                   |
|--------------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>GET</b>                                                                                                   | <b>/api/reports/my</b>            |
| Returns all reports belonging to the currently authenticated student. Results are sorted by date descending. |                                   |
| <b>Auth</b>                                                                                                  | Requires JWT token — Student only |

### Query Parameters (all optional)

| Field    | Type    | Required | Example    | Description                                                          |
|----------|---------|----------|------------|----------------------------------------------------------------------|
| status   | string  | No       | approved   | Filter by status: draft, pending, needs_revision, approved, rejected |
| fromDate | string  | No       | 2025-05-01 | Include only reports on or after this date (YYYY-MM-DD)              |
| toDate   | string  | No       | 2025-05-31 | Include only reports on or before this date (YYYY-MM-DD)             |
| page     | integer | No       | 1          | Page number for pagination                                           |
| limit    | integer | No       | 20         | Results per page                                                     |

## 6.4 Get My Report by ID (Student)

|                                                                                     |                                   |
|-------------------------------------------------------------------------------------|-----------------------------------|
| <b>GET</b>                                                                          | <b>/api/reports/my/:reportId</b>  |
| Returns a single report by ID. The report must belong to the authenticated student. |                                   |
| <b>Auth</b>                                                                         | Requires JWT token — Student only |

### URL Parameters

| Field    | Type   | Required | Example       | Description                 |
|----------|--------|----------|---------------|-----------------------------|
| reportId | string | Yes      | report-abc123 | The unique ID of the report |

## 6.5 List All Reports (Admin)

|                                                                                                                                                                      |                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| <b>GET</b>                                                                                                                                                           | <b>/api/reports</b>             |
| Returns all submitted reports across all students. Draft reports are excluded. Supports extensive filtering. Each result is enriched with basic student information. |                                 |
| <b>Auth</b>                                                                                                                                                          | Requires JWT token — Admin only |

### Query Parameters (all optional)

| Field | Type | Required | Example | Description |
|-------|------|----------|---------|-------------|
|-------|------|----------|---------|-------------|

|           |         |    |             |                                                                   |
|-----------|---------|----|-------------|-------------------------------------------------------------------|
| studentId | string  | No | student-001 | Filter by a specific student                                      |
| batchId   | string  | No | batch-001   | Filter by batch                                                   |
| courseId  | string  | No | course-001  | Filter by course                                                  |
| groupId   | string  | No | group-001   | Filter by group                                                   |
| status    | string  | No | pending     | Filter by report status                                           |
| fromDate  | string  | No | 2025-05-01  | Start of date range (YYYY-MM-DD)                                  |
| toDate    | string  | No | 2025-05-31  | End of date range (YYYY-MM-DD)                                    |
| isLate    | boolean | No | true        | Filter by whether the report was submitted late                   |
| minScore  | integer | No | 1           | Include only reports with evaluation score at or above this value |
| maxScore  | integer | No | 3           | Include only reports with evaluation score at or below this value |
| page      | integer | No | 1           | Page number                                                       |
| limit     | integer | No | 20          | Results per page                                                  |

### 6.6 Get Report by ID (Admin)

|                                                                                                          |                                 |
|----------------------------------------------------------------------------------------------------------|---------------------------------|
| GET                                                                                                      | /api/reports/:reportId          |
| Admin retrieves a specific report by its ID. Includes all evaluation data and the full revision history. |                                 |
| Auth                                                                                                     | Requires JWT token — Admin only |

URL Parameters

| Field    | Type   | Required | Example       | Description                 |
|----------|--------|----------|---------------|-----------------------------|
| reportId | string | Yes      | report-abc123 | The unique ID of the report |

### 6.7 Evaluate Report (Admin)

|                                                                                                                                                                                     |                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| PUT                                                                                                                                                                                 | /api/reports/:reportId/evaluate |
| Admin reviews and evaluates a submitted report. Status is the only required field. All evaluation fields can be updated at any time including changing a previously set evaluation. |                                 |
| Auth                                                                                                                                                                                | Requires JWT token — Admin only |

URL Parameters

| Field | Type | Required | Example | Description |
|-------|------|----------|---------|-------------|
|-------|------|----------|---------|-------------|

|          |        |     |               |                                         |
|----------|--------|-----|---------------|-----------------------------------------|
| reportId | string | Yes | report-abc123 | The unique ID of the report to evaluate |
|----------|--------|-----|---------------|-----------------------------------------|

**Request Body**

| Field        | Type    | Required | Example           | Description                                                 |
|--------------|---------|----------|-------------------|-------------------------------------------------------------|
| status       | string  | Yes      | approved          | Must be one of: pending, needs_revision, approved, rejected |
| feedback     | string  | No       | Great reflection. | Written feedback visible to the student                     |
| starRating   | integer | No       | 4                 | Overall quality rating. Integer between 1 and 5             |
| score        | integer | No       | 4                 | Numeric score. Integer between 1 and 5                      |
| attendance   | string  | No       | present           | Attendance for that day. One of: present, absent, late      |
| productivity | string  | No       | good              | Productivity rating. One of: low, average, good, excellent  |

**Evaluation Flow Notes**

When status is set to needs\_revision, the student is able to edit and resubmit their report. Once resubmitted, status returns to pending automatically and the revision history entry is recorded. When status is set to approved or rejected, the student can view the feedback but cannot edit the report. Attendance is also updated in the attendance records when provided.



## 7. Attendance Endpoints

### 7.1 Attendance Status Values

| Field   | Type   | Required | Example | Description                                                    |
|---------|--------|----------|---------|----------------------------------------------------------------|
| present | string |          |         | Student submitted a report before the daily deadline           |
| late    | string |          |         | Student submitted a report after the daily deadline            |
| absent  | string |          |         | Student did not submit a report. Must be set manually by admin |

### 7.2 Automatic Attendance

When a student submits a report for the current day, attendance is automatically recorded. Present is set if the submission is before the deadline. Late is set if the submission is after the deadline. Absent is never set automatically. An admin must set it manually. Automatic entries can be overridden by the admin at any time.

### 7.3 Get My Attendance (Student)

|                                                                                                                                                                    |                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>GET</b>                                                                                                                                                         | <b>/api/attendance/my</b>         |
| Returns the attendance records for the currently authenticated student. Filter by month and year for a monthly view, or by fromDate and toDate for a custom range. |                                   |
| <b>Auth</b>                                                                                                                                                        | Requires JWT token — Student only |

#### Query Parameters (all optional)

| Field    | Type    | Required | Example    | Description                                                      |
|----------|---------|----------|------------|------------------------------------------------------------------|
| month    | integer | No       | 5          | Month number (1-12). Use with year to get a monthly view         |
| year     | integer | No       | 2025       | Four-digit year. Use with month                                  |
| fromDate | string  | No       | 2025-05-01 | Start of custom date range. Overrides month and year if provided |
| toDate   | string  | No       | 2025-05-31 | End of custom date range                                         |

### 7.4 Get All Attendance (Admin)

|            |                        |
|------------|------------------------|
| <b>GET</b> | <b>/api/attendance</b> |
|------------|------------------------|

|                                                                                                                           |                                 |
|---------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| Returns attendance records for all students or a specific student. Supports the same date filters as the student version. |                                 |
| Auth                                                                                                                      | Requires JWT token — Admin only |

Query Parameters (all optional)

| Field     | Type    | Required | Example     | Description                                   |
|-----------|---------|----------|-------------|-----------------------------------------------|
| studentId | string  | No       | student-001 | Filter attendance for a specific student only |
| month     | integer | No       | 5           | Month number. Use with year                   |
| year      | integer | No       | 2025        | Four-digit year                               |
| fromDate  | string  | No       | 2025-05-01  | Start of custom range                         |
| toDate    | string  | No       | 2025-05-31  | End of custom range                           |

7.5 Set Attendance Manually (Admin)

|                                                                                                                                                                                                                                                                    |                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| POST                                                                                                                                                                                                                                                               | /api/attendance                 |
| Admin manually records or overrides an attendance entry for any student on any date. If an entry already exists for that student on that date it is overridden. Manual entries are flagged with isManual: true so they are distinguishable from automatic entries. |                                 |
| Auth                                                                                                                                                                                                                                                               | Requires JWT token — Admin only |

Request Body

| Field     | Type   | Required | Example     | Description                                              |
|-----------|--------|----------|-------------|----------------------------------------------------------|
| studentId | string | Yes      | student-001 | ID of the student to record attendance for               |
| date      | string | Yes      | 2025-05-18  | Date to record attendance for in YYYY-MM-DD format       |
| status    | string | Yes      | absent      | Attendance status. Must be one of: present, absent, late |

## 8. Tag Endpoints

*Tags are internal labels used by admins to categorize students. Tags are never visible to students. All tag endpoints are admin only.*

### 8.1 List Tags

|                                                                            |                                 |
|----------------------------------------------------------------------------|---------------------------------|
| <b>GET</b>                                                                 | <b>/api/tags</b>                |
| Returns all tags defined in the system. Tags are shared across all admins. |                                 |
| <b>Auth</b>                                                                | Requires JWT token — Admin only |

### 8.2 Create Tag

|                                                                              |                                 |
|------------------------------------------------------------------------------|---------------------------------|
| <b>POST</b>                                                                  | <b>/api/tags</b>                |
| Creates a new custom tag. Tag names must be unique (case-insensitive check). |                                 |
| <b>Auth</b>                                                                  | Requires JWT token — Admin only |

#### Request Body

| Field | Type   | Required | Example     | Description                                       |
|-------|--------|----------|-------------|---------------------------------------------------|
| name  | string | Yes      | Rising Star | Unique tag name. Case-insensitive duplicate check |

### 8.3 Rename Tag

|                                                                                                  |                                 |
|--------------------------------------------------------------------------------------------------|---------------------------------|
| <b>PUT</b>                                                                                       | <b>/api/tags/:tagId</b>         |
| Renames an existing tag. The rename applies everywhere the tag appears. New name must be unique. |                                 |
| <b>Auth</b>                                                                                      | Requires JWT token — Admin only |

#### URL Parameters

| Field | Type   | Required | Example | Description             |
|-------|--------|----------|---------|-------------------------|
| tagId | string | Yes      | tag-001 | ID of the tag to rename |

#### Request Body

| Field | Type   | Required | Example       | Description  |
|-------|--------|----------|---------------|--------------|
| name  | string | Yes      | Top Performer | New tag name |

### 8.4 Delete Tag

|               |                         |
|---------------|-------------------------|
| <b>DELETE</b> | <b>/api/tags/:tagId</b> |
|---------------|-------------------------|

Permanently deletes a tag. The tag is also removed from all students it was assigned to. This action cannot be undone.

**Auth** Requires JWT token — Admin only

#### URL Parameters

| Field | Type   | Required | Example | Description             |
|-------|--------|----------|---------|-------------------------|
| tagId | string | Yes      | tag-001 | ID of the tag to delete |

## 8.5 Assign Tag to Student

**POST** /api/tags/assign

Assigns an existing tag to a student. A student can have multiple tags. Assigning the same tag twice returns a 400 error.

**Auth** Requires JWT token — Admin only

#### Request Body

| Field     | Type   | Required | Example     | Description              |
|-----------|--------|----------|-------------|--------------------------|
| studentId | string | Yes      | student-001 | ID of the student to tag |
| tagId     | string | Yes      | tag-001     | ID of the tag to assign  |

## 8.6 Remove Tag from Student

**DELETE** /api/tags/assign/:studentId/:tagId

Removes a specific tag assignment from a student. The tag itself is not deleted.

**Auth** Requires JWT token — Admin only

#### URL Parameters

| Field     | Type   | Required | Example     | Description                              |
|-----------|--------|----------|-------------|------------------------------------------|
| studentId | string | Yes      | student-001 | ID of the student                        |
| tagId     | string | Yes      | tag-001     | ID of the tag to remove from the student |

## 8.7 Get Student Tags

**GET** /api/tags/student/:studentId

Returns all tags currently assigned to a specific student.

**Auth** Requires JWT token — Admin only

#### URL Parameters

| Field     | Type   | Required | Example     | Description                              |
|-----------|--------|----------|-------------|------------------------------------------|
| studentId | string | Yes      | student-001 | ID of the student whose tags to retrieve |

## 9. Analytics Endpoints

Analytics endpoints generate performance summaries based on report data, evaluation scores, and attendance. Summaries are calculated in real time on each request.

### 9.1 Performance Summary Fields

Every weekly and monthly summary includes the following data:

| Field                 | Type    | Required | Example | Description                                                                  |
|-----------------------|---------|----------|---------|------------------------------------------------------------------------------|
| reports.submitted     | integer |          |         | Total reports submitted in the period                                        |
| reports.onTime        | integer |          |         | Reports submitted before the daily deadline                                  |
| reports.late          | integer |          |         | Reports submitted after the deadline                                         |
| reports.approved      | integer |          |         | Reports approved by an admin                                                 |
| reports.rejected      | integer |          |         | Reports rejected by an admin                                                 |
| reports.needsRevision | integer |          |         | Reports currently pending revision                                           |
| reports.pending       | integer |          |         | Reports awaiting admin review                                                |
| completionRate        | string  |          |         | Percentage of expected daily reports submitted. e.g. 71.4%                   |
| scores.average        | string  |          |         | Average admin numeric score (1-5). Null if no scores exist                   |
| scores.count          | integer |          |         | Number of reports with a score in this period                                |
| ratings.average       | string  |          |         | Average admin star rating (1-5). Null if no ratings exist                    |
| productivity.average  | string  |          |         | Average productivity label: low, average, good, or excellent                 |
| attendance.present    | integer |          |         | Days marked present in the period                                            |
| attendance.absent     | integer |          |         | Days marked absent in the period                                             |
| attendance.late       | integer |          |         | Days marked late in the period                                               |
| feedback              | array   |          |         | Array of admin feedback entries for the period with reportId, date, and text |

## 9.2 Admin Dashboard

|                                                                                                                         |                                 |
|-------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| GET                                                                                                                     | /api/analytics/dashboard        |
| Returns a high-level overview of system activity for the admin dashboard. Shows items that require immediate attention. |                                 |
| Auth                                                                                                                    | Requires JWT token — Admin only |

### Success Response (200)

```
{
  "success": true,
  "data": {
    "pendingReports": 5,
    "needsRevision": 2,
    "missingReportsToday": [
      { "id": "student-002", "name": "Muhib ur Rehman", "groupId": "group-003" }
    ],
    "totalActiveStudents": 2,
    "submissionDeadline": "23:00"
  }
}
```

## 9.3 All Students Overview (Admin)

|                                                                                                                                                                                     |                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| GET                                                                                                                                                                                 | /api/analytics/students         |
| Returns a performance snapshot for every active student showing their current week and current month summary side by side. Supports filtering by batch, course, group, and sorting. |                                 |
| Auth                                                                                                                                                                                | Requires JWT token — Admin only |

### Query Parameters (all optional)

| Field    | Type    | Required | Example    | Description                                                   |
|----------|---------|----------|------------|---------------------------------------------------------------|
| batchId  | string  | No       | batch-001  | Filter to students in a specific batch                        |
| courseId | string  | No       | course-001 | Filter to students in a specific course                       |
| groupId  | string  | No       | group-001  | Filter to students in a specific group                        |
| sortBy   | string  | No       | name       | Sort order. One of: name, score, completion. Defaults to name |
| page     | integer | No       | 1          | Page number for pagination                                    |

|       |         |    |    |                  |
|-------|---------|----|----|------------------|
| limit | integer | No | 20 | Results per page |
|-------|---------|----|----|------------------|

## 9.4 Student Weekly Summary (Admin)

|                                                                                                                                                                       |                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| <b>GET</b>                                                                                                                                                            | <b>/api/analytics/students/:studentId/weekly</b> |
| Returns the weekly performance summary for a specific student. Use weekOffset to retrieve past weeks. 0 is the current Monday-to-Sunday week. 1 is the previous week. |                                                  |
| <b>Auth</b>                                                                                                                                                           | Requires JWT token — Admin only                  |

### URL Parameters

| Field     | Type   | Required | Example     | Description       |
|-----------|--------|----------|-------------|-------------------|
| studentId | string | Yes      | student-001 | ID of the student |

### Query Parameters

| Field      | Type    | Required | Example | Description                                                      |
|------------|---------|----------|---------|------------------------------------------------------------------|
| weekOffset | integer | No       | 0       | 0 is current week, 1 is last week, 2 is two weeks ago, and so on |

## 9.5 Student Monthly Summary (Admin)

|                                                                                                                       |                                                   |
|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <b>GET</b>                                                                                                            | <b>/api/analytics/students/:studentId/monthly</b> |
| Returns the monthly performance summary for a specific student. Also includes a week-by-week breakdown for the month. |                                                   |
| <b>Auth</b>                                                                                                           | Requires JWT token — Admin only                   |

### URL Parameters

| Field     | Type   | Required | Example     | Description       |
|-----------|--------|----------|-------------|-------------------|
| studentId | string | Yes      | student-001 | ID of the student |

### Query Parameters

| Field       | Type    | Required | Example | Description                                             |
|-------------|---------|----------|---------|---------------------------------------------------------|
| monthOffset | integer | No       | 0       | 0 is current calendar month, 1 is last month, and so on |

## 9.6 My Weekly Summary (Student)

|            |                                 |
|------------|---------------------------------|
| <b>GET</b> | <b>/api/analytics/my/weekly</b> |
|------------|---------------------------------|



Returns the weekly performance summary for the currently authenticated student. The student sees only their own data. Rankings and comparisons to other students are never included.

**Auth** Requires JWT token — Student only

#### Query Parameters

| Field      | Type    | Required | Example | Description                       |
|------------|---------|----------|---------|-----------------------------------|
| weekOffset | integer | No       | 0       | 0 is current week, 1 is last week |

## 9.7 My Monthly Summary (Student)

**GET** /api/analytics/my/monthly

Returns the monthly performance summary with weekly breakdown for the currently authenticated student.

**Auth** Requires JWT token — Student only

#### Query Parameters

| Field       | Type    | Required | Example | Description                         |
|-------------|---------|----------|---------|-------------------------------------|
| monthOffset | integer | No       | 0       | 0 is current month, 1 is last month |

## 10. Quick Reference

### 10.1 Complete Endpoint List

| Method | Endpoint                               | Auth Required | Role    |
|--------|----------------------------------------|---------------|---------|
| POST   | /api/auth/login                        | No            | All     |
| GET    | /api/auth/me                           | Yes           | All     |
| PUT    | /api/auth/change-password              | Yes           | All     |
| PUT    | /api/auth/reset-password/:studentId    | Yes           | Admin   |
| GET    | /api/settings                          | Yes           | All     |
| PUT    | /api/settings/deadline                 | Yes           | Admin   |
| GET    | /api/students                          | Yes           | Admin   |
| POST   | /api/students                          | Yes           | Admin   |
| GET    | /api/students/:studentId               | Yes           | Admin   |
| PUT    | /api/students/:studentId               | Yes           | Admin   |
| PATCH  | /api/students/:studentId/toggle-status | Yes           | Admin   |
| GET    | /api/program/batches                   | Yes           | Admin   |
| GET    | /api/program/batches/:batchId          | Yes           | Admin   |
| POST   | /api/program/batches                   | Yes           | Admin   |
| POST   | /api/program/batches/:batchId/courses  | Yes           | Admin   |
| GET    | /api/program/courses                   | Yes           | Admin   |
| POST   | /api/program/courses                   | Yes           | Admin   |
| GET    | /api/program/groups                    | Yes           | Admin   |
| POST   | /api/program/groups                    | Yes           | Admin   |
| POST   | /api/reports                           | Yes           | Student |
| GET    | /api/reports/my                        | Yes           | Student |
| GET    | /api/reports/my/:reportId              | Yes           | Student |
| GET    | /api/reports                           | Yes           | Admin   |
| GET    | /api/reports/:reportId                 | Yes           | Admin   |
| PUT    | /api/reports/:reportId/evaluate        | Yes           | Admin   |
| GET    | /api/attendance/my                     | Yes           | Student |
| GET    | /api/attendance                        | Yes           | Admin   |
| POST   | /api/attendance                        | Yes           | Admin   |
| GET    | /api/tags                              | Yes           | Admin   |
| POST   | /api/tags                              | Yes           | Admin   |

|               |                                            |     |         |
|---------------|--------------------------------------------|-----|---------|
| <b>PUT</b>    | /api/tags/:tagId                           | Yes | Admin   |
| <b>DELETE</b> | /api/tags/:tagId                           | Yes | Admin   |
| <b>POST</b>   | /api/tags/assign                           | Yes | Admin   |
| <b>DELETE</b> | /api/tags/assign/:studentId/:tagId         | Yes | Admin   |
| <b>GET</b>    | /api/tags/student/:studentId               | Yes | Admin   |
| <b>GET</b>    | /api/analytics/dashboard                   | Yes | Admin   |
| <b>GET</b>    | /api/analytics/students                    | Yes | Admin   |
| <b>GET</b>    | /api/analytics/students/:studentId/weekly  | Yes | Admin   |
| <b>GET</b>    | /api/analytics/students/:studentId/monthly | Yes | Admin   |
| <b>GET</b>    | /api/analytics/my/weekly                   | Yes | Student |
| <b>GET</b>    | /api/analytics/my/monthly                  | Yes | Student |

## 10.2 Important Notes

| Field        | Type                                                                                                                                                                                     | Required | Example | Description |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------|-------------|
| JSON Storage | All data is read from and written to src/data/*.json files. The files are overwritten on every write operation. Always back up the data directory before testing destructive operations. |          |         |             |
| Token Expiry | JWT tokens expire after 7 days. After expiry, the user must log in again to receive a new token.                                                                                         |          |         |             |
| Pagination   | All list endpoints support page and limit query parameters. Default is page 1 with a limit of 20 results.                                                                                |          |         |             |
| Date Format  | All dates must be provided in YYYY-MM-DD format. Report dates cannot be in the future. Late submissions are                                                                              |          |         |             |

|                  |                                                                                                                                                |  |  |  |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|
|                  | limited to 3 calendar days in the past.                                                                                                        |  |  |  |
| Score and Rating | Admin scores and star ratings both use an integer scale of 1 to 5. Non-integer values are rejected.                                            |  |  |  |
| CORS             | CORS is enabled with default settings allowing all origins. Restrict this before deploying to any environment beyond localhost.                |  |  |  |
| Migration Path   | All file reads and writes are centralized in src/Utils/storage.js. Migrating to a database only requires replacing the functions in that file. |  |  |  |